Unit 37: Integration by Parts — Full Solutions

Warm-up

1. $\displaystyle\int x\cos x\,dx$

Let $u = x$ $(du = dx)$, $dv = \cos x\,dx$ $(v = \sin x)$.

$$\int x\cos x\,dx = x\sin x - \int \sin x\,dx = x\sin x - (-\cos x) = x\sin x + \cos x + C$$

2. $\displaystyle\int xe^{2x}\,dx$

Let $u = x$ $(du = dx)$, $dv = e^{2x}\,dx$ $(v = \frac{1}{2}e^{2x})$.

$$\int xe^{2x}\,dx = \frac{x}{2}e^{2x} - \int \frac{1}{2}e^{2x}\,dx = \frac{x}{2}e^{2x} - \frac{1}{4}e^{2x} + C$$

3. $\displaystyle\int x\sin(2x)\,dx$

Let $u = x$ $(du = dx)$, $dv = \sin(2x)\,dx$ $(v = -\frac{1}{2}\cos(2x))$.

$$\int x\sin(2x)\,dx = -\frac{x}{2}\cos(2x) - \int \left(-\frac{1}{2}\cos(2x)\right)dx = -\frac{x}{2}\cos(2x) + \frac{1}{2}\int \cos(2x)\,dx$$

$$= -\frac{x}{2}\cos(2x) + \frac{1}{4}\sin(2x) + C$$

4. $\displaystyle\int \ln x\,dx$

Let $u = \ln x$ $(du = \frac{1}{x}\,dx)$, $dv = dx$ $(v = x)$.

$$\int \ln x\,dx = x\ln x - \int x\cdot\frac{1}{x}\,dx = x\ln x - \int 1\,dx = x\ln x - x + C$$

5. $\displaystyle\int (x+1)e^x\,dx$

Let $u = x+1$ $(du = dx)$, $dv = e^x\,dx$ $(v = e^x)$.

$$\int (x+1)e^x\,dx = (x+1)e^x - \int e^x\,dx = (x+1)e^x - e^x + C = xe^x + e^x - e^x + C = xe^x + C$$

6. $\displaystyle\int xe^{-x}\,dx$

Let $u = x$ $(du = dx)$, $dv = e^{-x}\,dx$ $(v = -e^{-x})$.

$$\int xe^{-x}\,dx = -xe^{-x} - \int(-e^{-x})\,dx = -xe^{-x} + \int e^{-x}\,dx = -xe^{-x} - e^{-x} + C = -e^{-x}(x+1) + C$$

Standard

7. $\displaystyle\int x\sin x\,dx$

Let $u = x$ $(du = dx)$, $dv = \sin x\,dx$ $(v = -\cos x)$.

$$\int x\sin x\,dx = -x\cos x - \int(-\cos x)\,dx = -x\cos x + \int\cos x\,dx = -x\cos x + \sin x + C$$

8. $\displaystyle\int x^5\ln x\,dx$

Let $u = \ln x$ $(du = \frac{1}{x}\,dx)$, $dv = x^5\,dx$ $(v = \frac{x^6}{6})$.

$$\int x^5\ln x\,dx = \frac{x^6}{6}\ln x - \int\frac{x^6}{6}\cdot\frac{1}{x}\,dx = \frac{x^6}{6}\ln x - \frac{1}{6}\int x^5\,dx$$

$$= \frac{x^6}{6}\ln x - \frac{1}{6}\cdot\frac{x^6}{6} + C = \frac{x^6}{6}\ln x - \frac{x^6}{36} + C$$

9. $\displaystyle\int x^2e^x\,dx$

Round 1: $u = x^2$ $(du = 2x\,dx)$, $dv = e^x\,dx$ $(v = e^x)$.

$$\int x^2e^x\,dx = x^2e^x - \int 2xe^x\,dx = x^2e^x - 2\int xe^x\,dx$$

Round 2: we need $\displaystyle\int xe^x\,dx$. Let $u = x$ $(du = dx)$, $dv = e^x\,dx$ $(v = e^x)$.

$$\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C$$

Combine:

$$\int x^2e^x\,dx = x^2e^x - 2(xe^x - e^x) + C = x^2e^x - 2xe^x + 2e^x + C = e^x(x^2 - 2x + 2) + C$$

10. $\displaystyle\int \ln(x^2)\,dx$

Simplify first: $\ln(x^2) = 2\ln x$.

$$\int 2\ln x\,dx = 2\int \ln x\,dx = 2(x\ln x - x) + C = 2x\ln x - 2x + C$$

11. $\int x\cos(3x)\,dx$

Let $u = x$ $(du = dx)$, $dv = \cos(3x)\,dx$ $(v = \frac{1}{3}\sin(3x))$.

$$\int x\cos(3x)\,dx = \frac{x}{3}\sin(3x) - \int \frac{1}{3}\sin(3x)\,dx = \frac{x}{3}\sin(3x) - \frac{1}{3}\left(-\frac{1}{3}\cos(3x)\right) + C$$

$$= \frac{x}{3}\sin(3x) + \frac{1}{9}\cos(3x) + C$$

12. $\int \arctan(x)\,dx$

Let $u = \arctan x$ $(du = \frac{1}{1+x^2}\,dx)$, $dv = dx$ $(v = x)$.

$$\int \arctan(x)\,dx = x\arctan x - \int \frac{x}{1+x^2}\,dx$$

For the remaining integral, let $w = 1 + x^2$, $dw = 2x\,dx$:

$$\int \frac{x}{1+x^2}\,dx = \frac{1}{2}\int \frac{dw}{w} = \frac{1}{2}\ln(1+x^2)$$

Combine:

$$\int \arctan(x)\,dx = x\arctan x - \frac{1}{2}\ln(1+x^2) + C$$

13. $\int x^3\ln x\,dx$

Let $u = \ln x$ $(du = \frac{1}{x}\,dx)$, $dv = x^3\,dx$ $(v = \frac{x^4}{4})$.

$$\int x^3\ln x\,dx = \frac{x^4}{4}\ln x - \int \frac{x^4}{4}\cdot\frac{1}{x}\,dx = \frac{x^4}{4}\ln x - \frac{1}{4}\int x^3\,dx$$

$$= \frac{x^4}{4}\ln x - \frac{1}{4}\cdot\frac{x^4}{4} + C = \frac{x^4}{4}\ln x - \frac{x^4}{16} + C$$

Challenge

14. $\int xe^x \, dx$

Let $u = x$ $(du = dx)$, $dv = e^x \, dx$ $(v = e^x)$.

$$\int xe^x \, dx = xe^x - \int e^x \, dx = xe^x - e^x + C$$

15. $\int e^x \cos x \, dx$

Let $I = \int e^x \cos x \, dx$.

Round 1: $u = e^x$ $(du = e^x \, dx)$, $dv = \cos x \, dx$ $(v = \sin x)$.

$$I = e^x \sin x - \int e^x \sin x \, dx$$

Round 2: for $\int e^x \sin x \, dx$, let $u = e^x$ $(du = e^x \, dx)$, $dv = \sin x \, dx$ $(v = -\cos x)$.

$$\int e^x \sin x \, dx = -e^x \cos x - \int (-\cos x)(e^x \, dx) = -e^x \cos x + \int e^x \cos x \, dx = -e^x \cos x + I$$

Substitute back into the Round 1 result:

$$I = e^x \sin x - \left(-e^x \cos x + I \right) = e^x \sin x + e^x \cos x - I$$

Solve for I algebraically:

$$I + I = e^x \sin x + e^x \cos x$$

$$2I = e^x (\sin x + \cos x)$$

$$I = \frac{e^x (\sin x + \cos x)}{2} + C$$

16. $\int e^{2x} \sin x \, dx$

Let $I = \int e^{2x} \sin x \, dx$.

Round 1: $u = e^{2x}$ $(du = 2e^{2x} \, dx)$, $dv = \sin x \, dx$ $(v = -\cos x)$.

$$I = -e^{2x} \cos x + \int 2e^{2x} \cos x \, dx = -e^{2x} \cos x + 2 \int e^{2x} \cos x \, dx$$

Round 2: let $J = \int e^{2x} \cos x \, dx$. Using $u = e^{2x}$ $(du = 2e^{2x}\, dx)$, $dv = \cos x \, dx$ $(v = \sin x)$:

$$J = e^{2x} \sin x - \int 2e^{2x} \sin x \, dx = e^{2x} \sin x - 2I$$

Substitute back:

$$I = -e^{2x} \cos x + 2(e^{2x} \sin x - 2I) = -e^{2x} \cos x + 2e^{2x} \sin x - 4I$$

Solve for I:

$$I + 4I = -e^{2x} \cos x + 2e^{2x} \sin x$$

$$5I = e^{2x}(2 \sin x - \cos x)$$

$$I = \frac{e^{2x}(2 \sin x - \cos x)}{5} + C$$

17. $\int x^2 \sin x \, dx$

Round 1: $u = x^2$ $(du = 2x\, dx)$, $dv = \sin x \, dx$ $(v = -\cos x)$.

$$\int x^2 \sin x \, dx = -x^2 \cos x - \int (-\cos x)(2x\, dx) = -x^2 \cos x + 2 \int x \cos x \, dx$$

Round 2: from Problem 1, $\int x \cos x \, dx = x \sin x + \cos x + C$.

Combine:

$$\int x^2 \sin x \, dx = -x^2 \cos x + 2(x \sin x + \cos x) + C = -x^2 \cos x + 2x \sin x + 2 \cos x + C$$

18. $\int x(\ln x)^2 \, dx$

Round 1: let $u = (\ln x)^2$ $(du = 2 \ln x \cdot \frac{1}{x}\, dx)$, $dv = x\, dx$ $(v = \frac{x^2}{2})$.

$$\int x(\ln x)^2 \, dx = \frac{x^2}{2}(\ln x)^2 - \int \frac{x^2}{2} \cdot \frac{2 \ln x}{x} \, dx = \frac{x^2}{2}(\ln x)^2 - \int x \ln x \, dx$$

Round 2: for $\int x \ln x \, dx$, let $u = \ln x$ $(du = \frac{1}{x}\, dx)$, $dv = x\, dx$ $(v = \frac{x^2}{2})$.

$$\int x \ln x \, dx = \frac{x^2}{2} \ln x - \int \frac{x^2}{2} \cdot \frac{1}{x} \, dx = \frac{x^2}{2} \ln x - \frac{1}{2} \int x \, dx = \frac{x^2}{2} \ln x - \frac{x^2}{4}$$

Combine:

$$\int x(\ln x)^2\,dx = \underline{\frac{x^2}{2}(\ln x)^2 - \left(\frac{x^2}{2}\ln x - \frac{x^2}{4}\right) + C = \frac{x^2}{2}(\ln x)^2 - \frac{x^2}{2}\ln x} + \frac{x^2}{4} + C$$

Applied

19. $\displaystyle\int_0^{\pi} x\sin x\,dx$

Using the antiderivative from Problem 7: $-x\cos x + \sin x$.

$$\Big[-x\cos x + \sin x\Big]_0^{\pi} = (-\pi\cos\pi + \sin\pi) - (0+0) = -\pi(-1) + 0 = \pi$$

Answer: total work done $= \pi$ (Newton-meters).

20. $\displaystyle\int_0^{10} te^{-t}\,dt$

Using the antiderivative pattern from Problem 6: $-e^{-t}(t+1)$.

$$\Big[-e^{-t}(t+1)\Big]_0^{10} = -e^{-10}(11) - (-e^{0}(1)) = -11e^{-10} + 1 = 1 - 11e^{-10}$$

Since e^{-10} is extremely small (≈ 0.0000454):

$$\approx 1 - 0.0005 \approx 0.9995$$

Answer: total present value $= 1 - 11e^{-10} \approx 0.9995$.

21. $\displaystyle\int_1^{e} x\ln x\,dx$

Using the antiderivative from Unit's earlier work (Problem 18's Round 2): $\dfrac{x^2}{2}\ln x - \dfrac{x^2}{4}$.

At $x = e$: $\dfrac{e^2}{2}(1) - \dfrac{e^2}{4} = \dfrac{e^2}{2} - \dfrac{e^2}{4} = \dfrac{e^2}{4}$.

At $x = 1$: $\dfrac{1}{2}(0) - \dfrac{1}{4} = -\dfrac{1}{4}$.

$$\frac{e^2}{4} - \left(-\frac{1}{4}\right) = \frac{e^2+1}{4}$$

Answer: $\dfrac{e^2+1}{4} \approx 2.097$.

22. $\displaystyle\int_0^{\pi/2} x\cos x\,dx$

Using the antiderivative from Problem 1: $x\sin x+\cos x$.

$$[x\sin x+\cos x]_0^{\pi/2}=\left(\frac{\pi}{2}\cdot 1+0\right)-(0+1)=\frac{\pi}{2}-1$$

Answer: $\frac{\pi}{2}-1\approx 0.571$.